IMAGE CLASSIFICATION ON CALTECH-UCSD BIRDS-200-2011

Course: Machine Learning

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13/10/2026

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EXPERIMENTS AND EVALUATION

CONCLUSION AND PERSPECTIVES



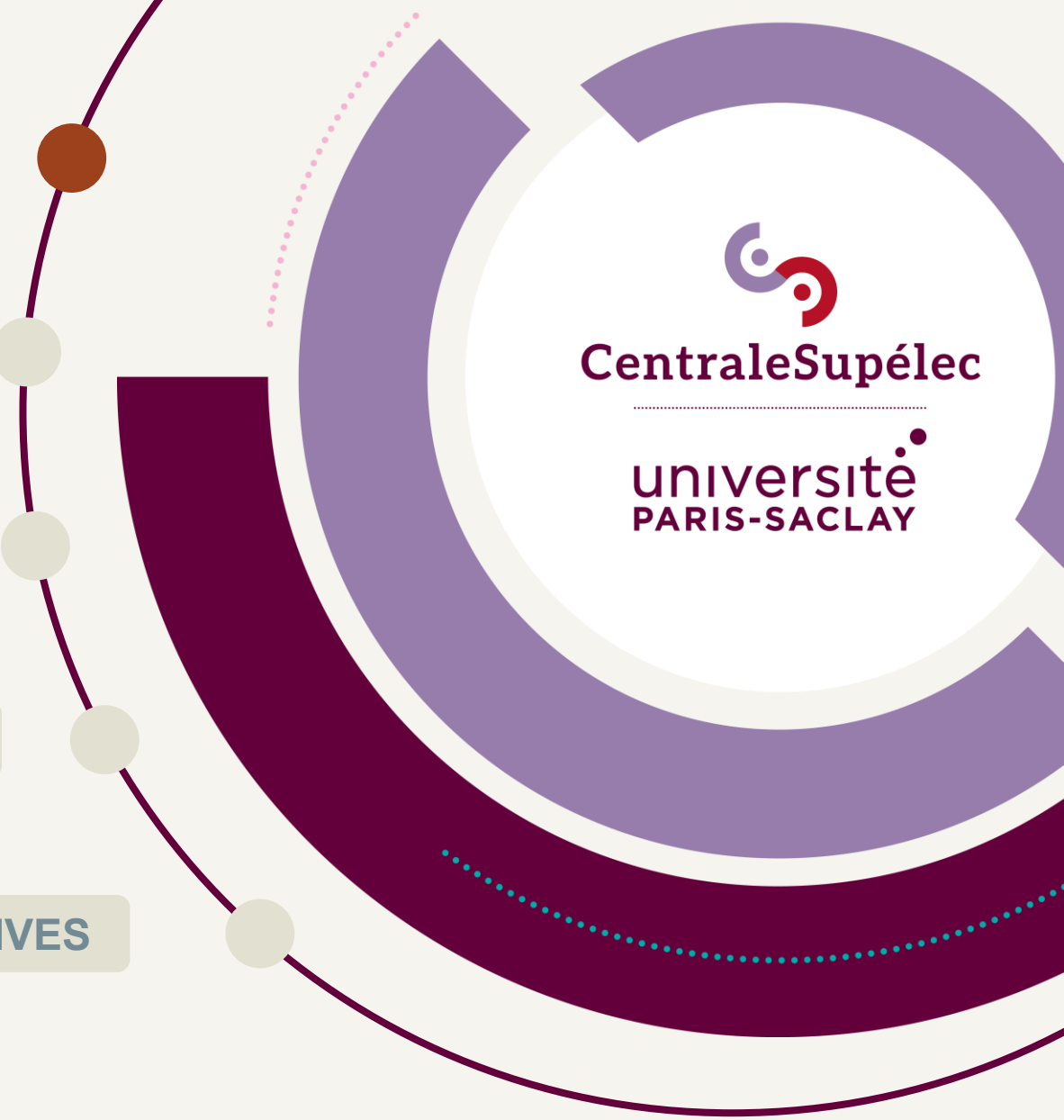
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INTRODUCTION TASK

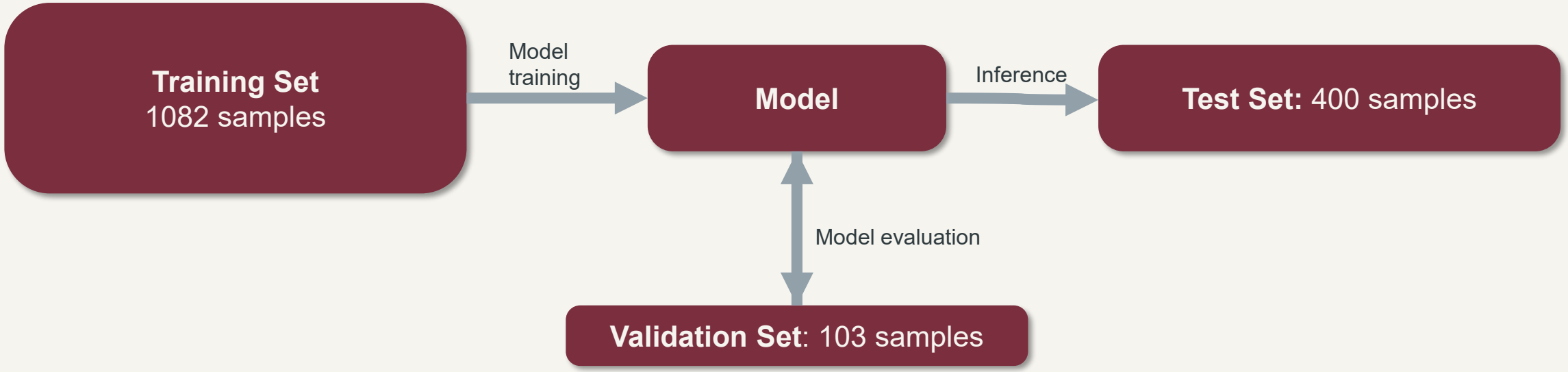
Task: Classify **20 bird species** from images (fine-grained recognition) on a Kaggle competition



Core difficulty:

- Fine-grained differences are subtle
- Huge nuisance variation: pose, scale, lighting, occlusion, background clutter
- Small data per class → overfitting risk + validation split noise
- Increased difficulty for near-identical species (e.g. **American vs Fish Crow**)

DATASET



Technical details:

- ✓ 51–58 images per class in training
- ✓ 2-8 images per class in validation
- ✓ Pre-resized images with a maximum side of 500 pixels

CLASSES

Bird species: variety of species regarding color and size



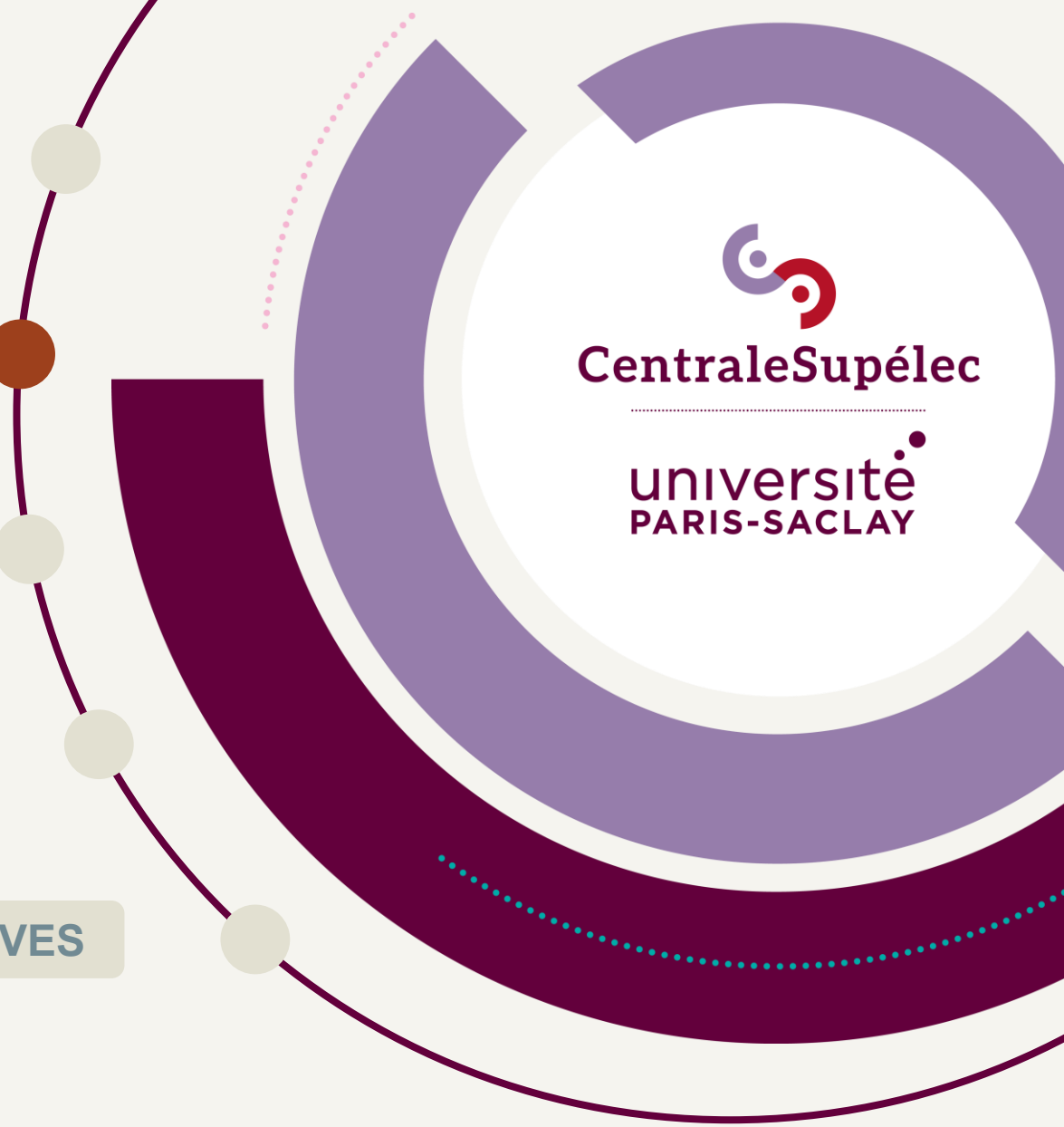
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BACKGROUND BIRD CLASSIFICATION



Classical era: hand-crafted local descriptors (e.g. HOG) with pooling (e.g. Fisher Vectors) and linear classifiers



Modern standard: **transfer learning** with pretrained CNNs/Transformers, lightweight head or fine-tuning



Typical winning tricks in low-data: cropping/segmentation to reduce background shortcuts, higher resolution, augmentation and cross-validation for reliable model selection

BACKBONE FAMILIES

CNNs

E.g. ConvNeXt/ConvNeXt-V2

Strong **local inductive bias** → great for textures/edges/feather detail

Robust when objects are partially visible

Vision Transformers

E.g. ViT, EVA-02

Model **global relationships** via self-attention

Benefit more from higher resolution for fine cues

HYBRID

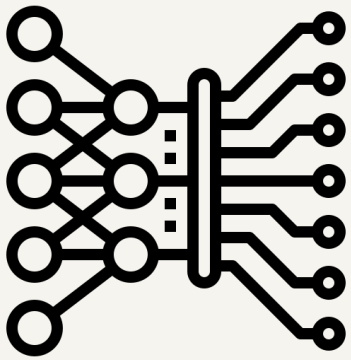
E.g. CAFormer-B36

Early conv blocks (local) to extract low-level features

Later attention (global) to to capture global semantic relationships

STATE OF THE ART BACKBONES

Access: **timm** of PyTorch to experiment with a wide range of SOTA pretrained backbones



Backbones explored

Baselines (old): ResNet-50, EfficientNet-B0

Modern CNNs: ConvNeXt, ConvNeXt-V2

Hybrid: CAFormer-B36

Transformers: ViT-SO150M (384), EVA-02 Large (448)

Frozen backbones due to small data

Regularization & efficiency

QUALITATIVE INSPECTION

Mistakes in fine-grained birds:

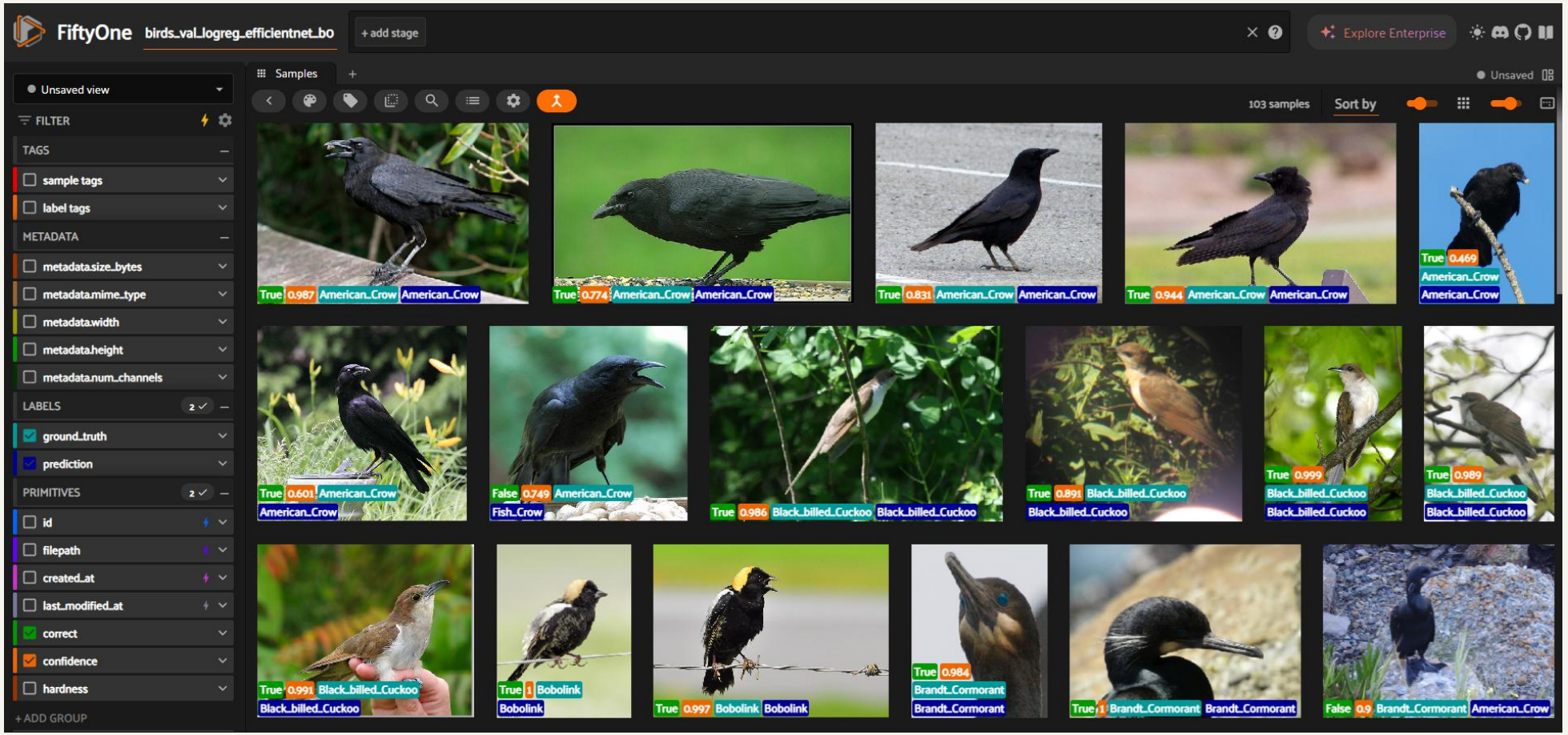
- A small set of visually similar birds can **dominate** the mistakes (American vs Fish crow)
- Often driven by **background bias**, **occlusion**, **pose**, or **scale**

Benefits:

- **Filter** instantly to only wrong predictions
- **Slice** by species to detect systematic confusions
- Sort by **low confidence** to surface ambiguous cases
- Visually check whether errors come from background bias, occlusion, pose, or viewpoint

QUALITATIVE INSPECTION

FiftyOne App to inspect validation predictions



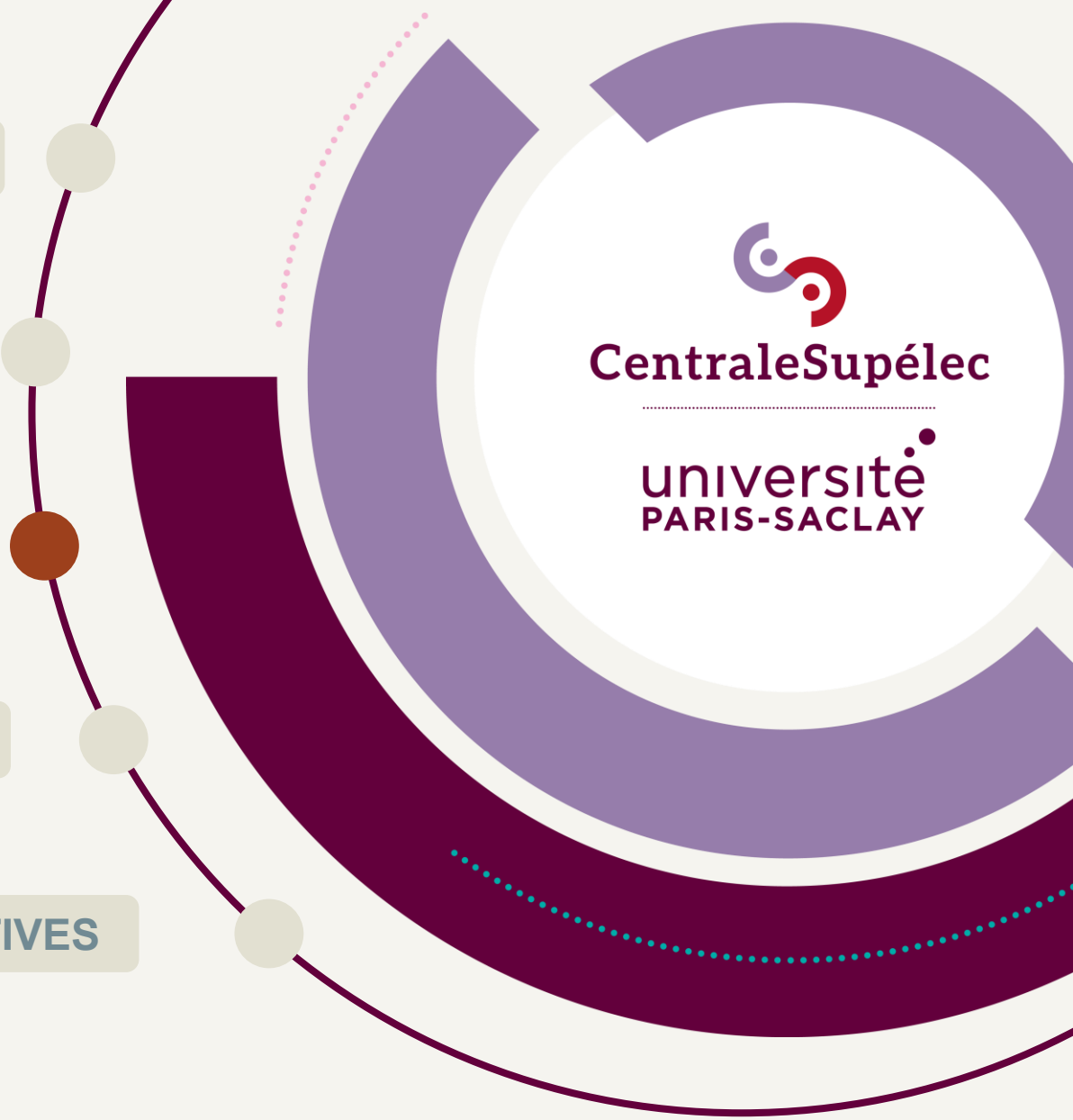
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FROZEN BACKBONES

Why?

Limited size of the training dataset (20 classes, 51-58 images in each, 1082 in total)
Fine-tuning the entire parameter space of large models is **expensive** and **prone to overfitting**

How?

TIMM (PyTorch Image Models) library **with State-of-the-Art pretrained backbones**
Transfer Learning, models **pretrained on large datasets** (initialized with num_classes=0)

Which ones?

Model Name	Resolution	Params (M)	Top-1 Acc (%)	Pretrain Data	Architecture	Key Feature
efficientnet_b0	224^2	5.3	77.70	IN-1k	CNN	Compound scaling baseline
resnet50	224^2	25.6	81.22	IN-1k	CNN	Residual connections
convnext_base	224^2	88.6	86.30	IN-22k	CNN	Modernized CNN design
convnextv2_base	384^2	88.7	87.64	IN-22k	CNN	Fully conv. MAE + FCMAE
vit_so150m2	384^2	136.3	87.93	IN-12k	ViT	Regularized ViT
caformer_b36	384^2	98.8	88.06	IN-22k	Hybrid	Conv + attention MetaFormer
convnextv2_large	384^2	198.0	88.18	IN-22k	CNN	Fully conv. MAE + FCMAE
eva02_large	448^2	305.1	90.06	M-38M	ViT	MIM with CLIP targets (SOTA)

HEAD EXPERIMENTS

Simple Linear

$$\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$$

State-of-the-art are **pre-trained on massive datasets**

They output latent spaces that are **highly linearly separable**

Keeps the **geometric relation between feature vectors**

Simple, efficient, and fast to train

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Multi-Layer Perceptron

$$\mathbf{y} = \text{Linear}(\text{Dropout}(\text{GELU}(\text{BatchNorm}(\text{Linear}(\mathbf{x})))))$$

Introduces non-linearity before the final classification

Frozen backbone features **might not be linearly separable**?

GeLU (ConvNeXt, …) **SiLU** (EfficientNet-B0) **RELU** (ResNet-50)

Trainable layers on top of backbones **made optimization harder**

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ArcFace AAM

Replace dot product with **cosine similarity** by **normalizing** both the **feature vectors** and the **class weights**

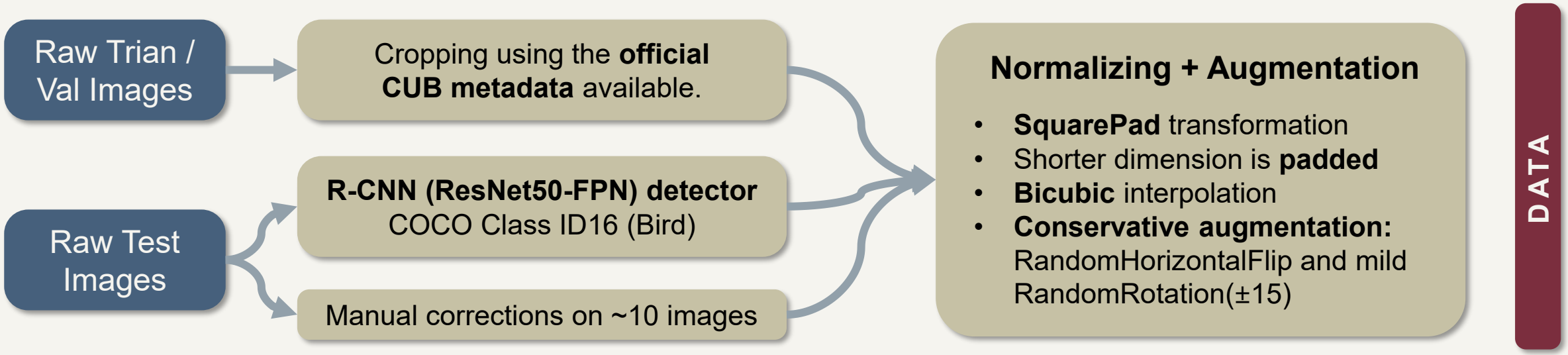
$$\cos(\theta_j) = \frac{\mathbf{W}_j^T \mathbf{x}_i}{\|\mathbf{W}_j\| \|\mathbf{x}_i\|}$$

In training, **additive angular margin** is added to the angle of the GT class (then cos. sim.)

$$L_{\text{Arc}} = -\log \frac{e^{s\cos(\theta_{y_i}+m)}}{e^{s\cos(\theta_{y_i}+m)} + \sum_{j \neq y_i} e^{s\cos\theta_j}}$$

Goal: **clearer decision boundaries** solve **high inter-class similarity**

OTHER ADDITIONS



OTHER ADDITIONS

Raw Trian / Val Images

Raw Test Images

Cropping using the **official CUB metadata** available.

R-CNN (ResNet50-FPN) detector
COCO Class ID16 (Bird)

Manual corrections on ~10 images

Normalizing + Augmentation

- **SquarePad** transformation
- Shorter dimension is **padded**
- **Bicubic** interpolation
- **Conservative augmentation:** RandomHorizontalFlip and mild RandomRotation(±15)

DATA

Optimizer and Weight Decay

AdamW optimizer
Initial **learning rate** of **1e-3**
Weight decay of **1e-4**
Cosine annealing **learning rate scheduler**

Better overall regularization

Loss and Label Smoothing

Label smoothing with a factor of α=0.1

Used as **high inter-class similarity** in our task
Assigns a target *p* of 0.9 to the GT class and distributes the remaining uniformly.

Better overall regularization

MODEL

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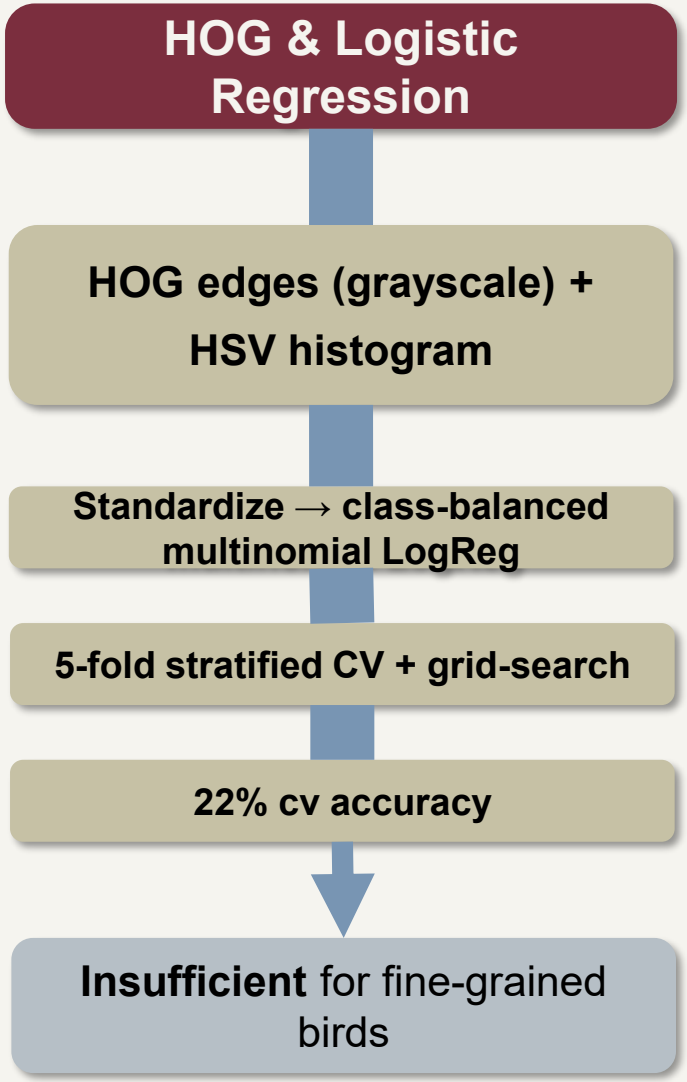
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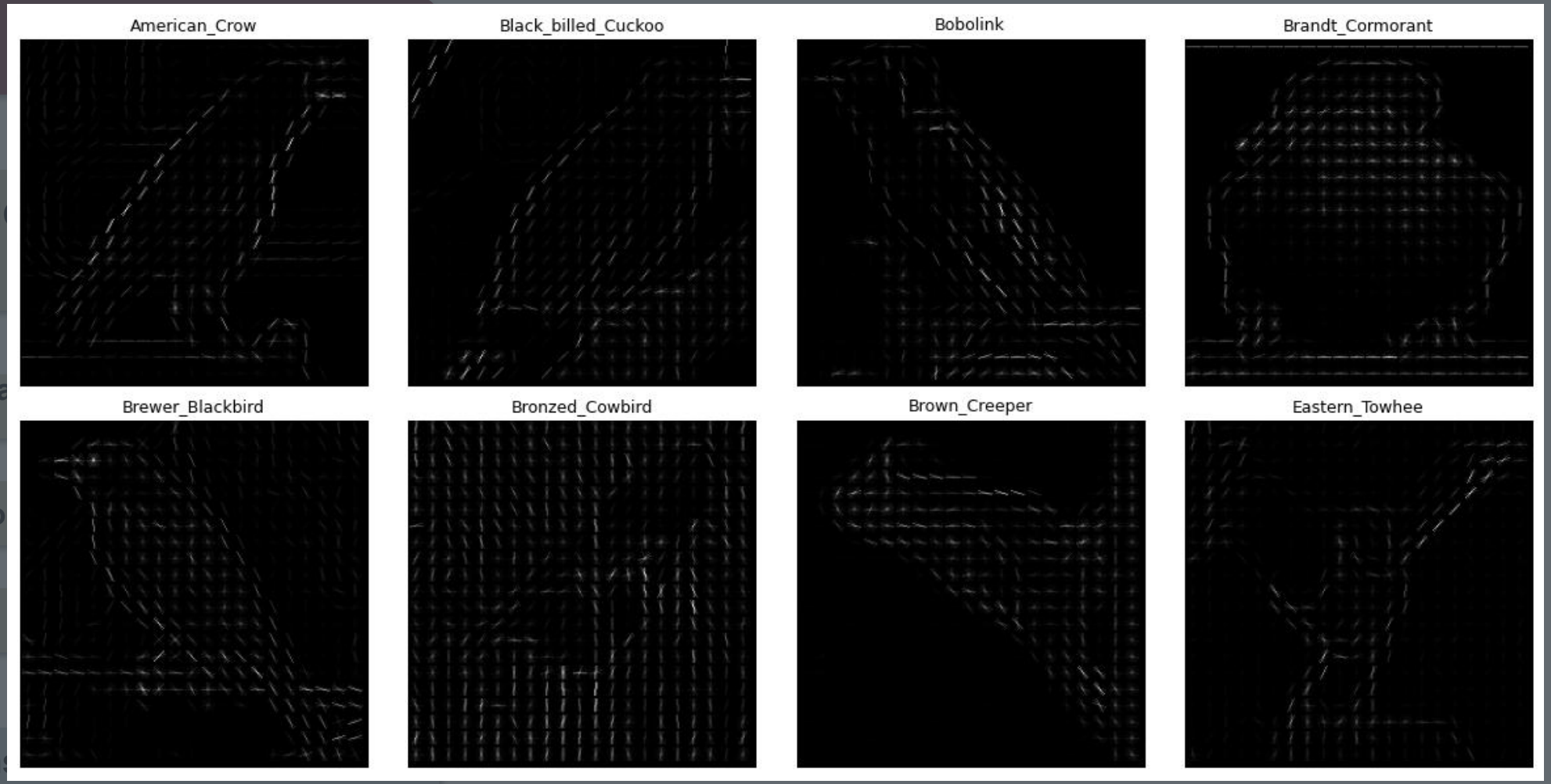
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CLASSICAL APPROACHES



CLASSICAL APPROACHES



birds

CLASSICAL APPROACHES

HOG & Logistic Regression

HOG edges (grayscale) + HSV histogram

Standardize → class-balanced multinomial LogReg

5-fold stratified CV + grid-search

22% cv accuracy

Insufficient for fine-grained birds

Fisher Vectors + Linear SVM

Dense SIFT → GMM → Improved FV

Class-balanced linear SVM

5-fold stratified CV

38.5% cv accuracy

t-SNE shows **strong class mixing**

CLASSICAL APPROACHES

HOG & Logistic Regression

H

HSV histogram

Standardize → class-balanced

5-fo

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t-SNE shows strong class mixing

Capture generic texture/shape but fail on localized markings + pose variation



Focus on pretrained transfer learning

FAILED EXPERIMENTS – COMBINING MODELS & UNFREEZING

Hierarchical "Mixture of Experts"

Problem: **confusion between visually similar species**, such as American vs. Fish Crows

Two **specialized ConvNeXt V2 models** just on the **binary classification tasks** for Crows and Black birds

Inference: **General Model (ViT) predicts others**

Performance is **limited by data**, **not model architecture!**

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Ensemble of Our Best Models

Different architectures (Transformers vs. CNNs) would make **uncorrelated errors**?

EVA-02 Large (Arcface, 384)

ConvNeXt V2 Large (Linear, 384)

ViT-SO400M (Linear, 384)

Aggregate the **softmax probability distributions**

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Aggregate the **softmax probability distributions**

Partial Unfreezing of Backbones

Goal: avoid the **compute and overfitting risks** of full fine tuning

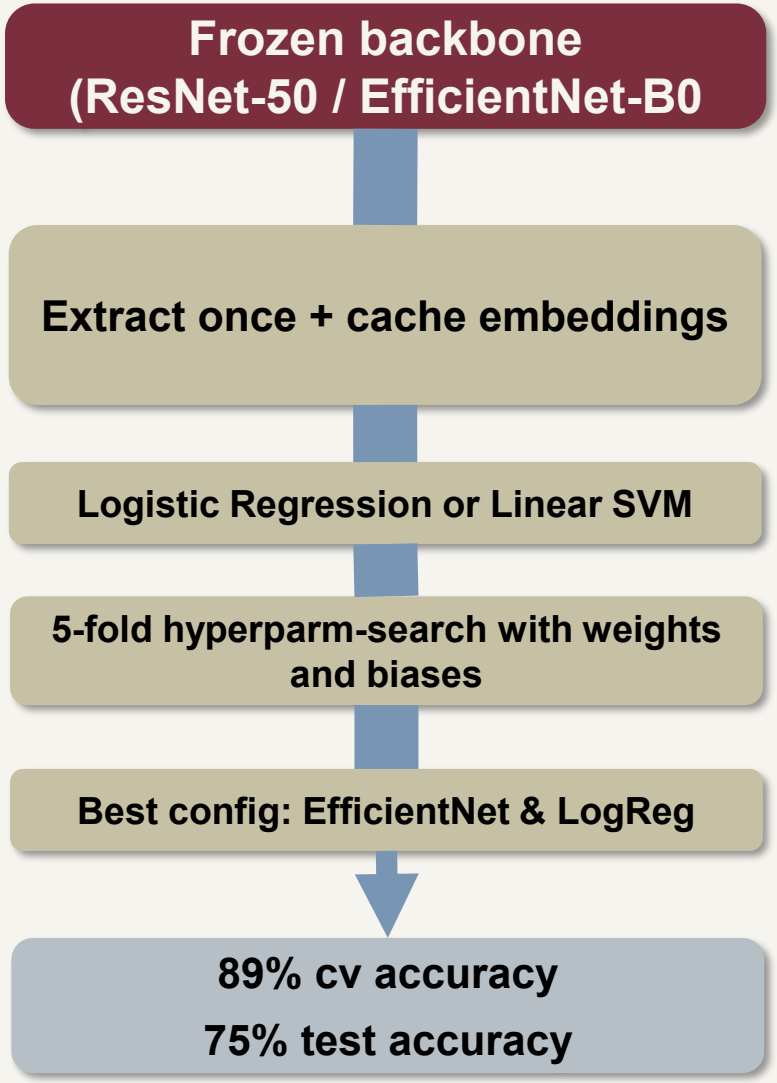
Later layers **learn more task-specific features**, unfreeze only the **last x%** of the backbone.

These layers are trained with a **differential learning rate**

$(\eta_{backbone} = 1e^{-5} \text{ vs. } \eta_{head} = 1e^{-3})$

Performance is **limited by data, not model architecture!**

PRETRAINED EMBEDDINGS & LINEAR CLASSIFIERS



PRETRAINED EMBEDDINGS - LINEAR CLASSIFIERS

Frozen backbone (ResNet-50 / EfficientNet-B0

Extract once + cache embeddings

Logistic Regression or Linear SVM

5-fold hyperparm-search with weights and biases

Best config: EfficientNet & LogReg

89% cv accuracy

75% test accuracy

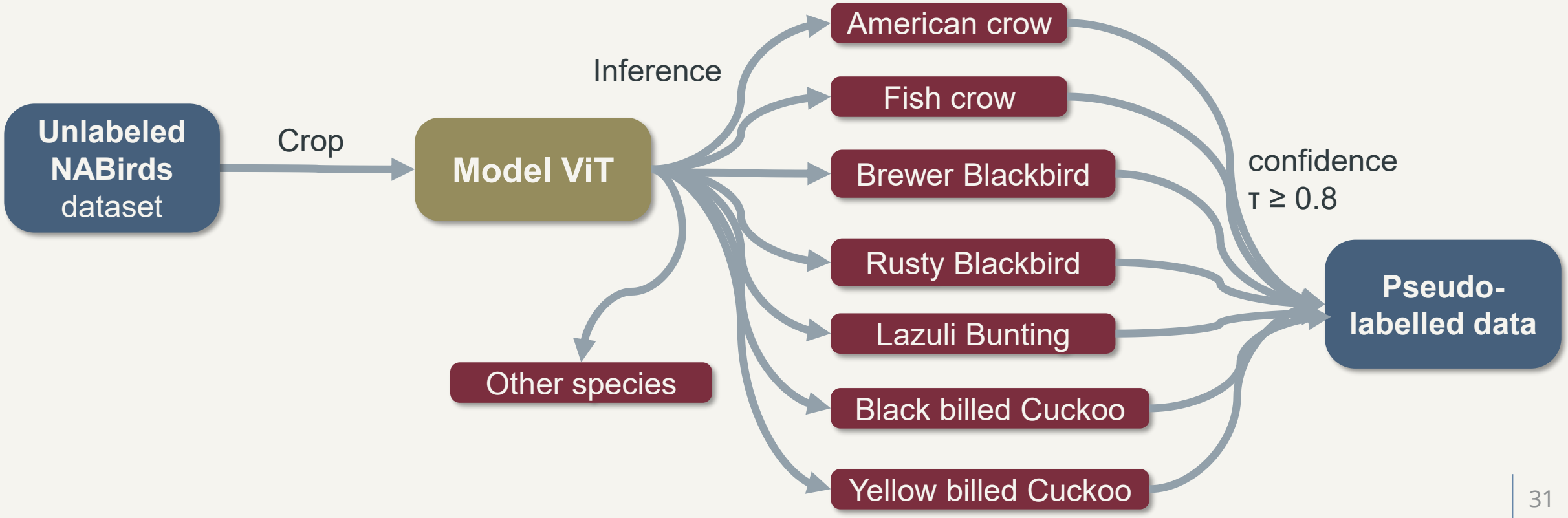


- ✓ **Best accuracy-compute tradeoff**
- ✓ **Stable + easy to tune**
- ✓ **Confidence scores - probabilities for analysis**

PSEUDO LABELING

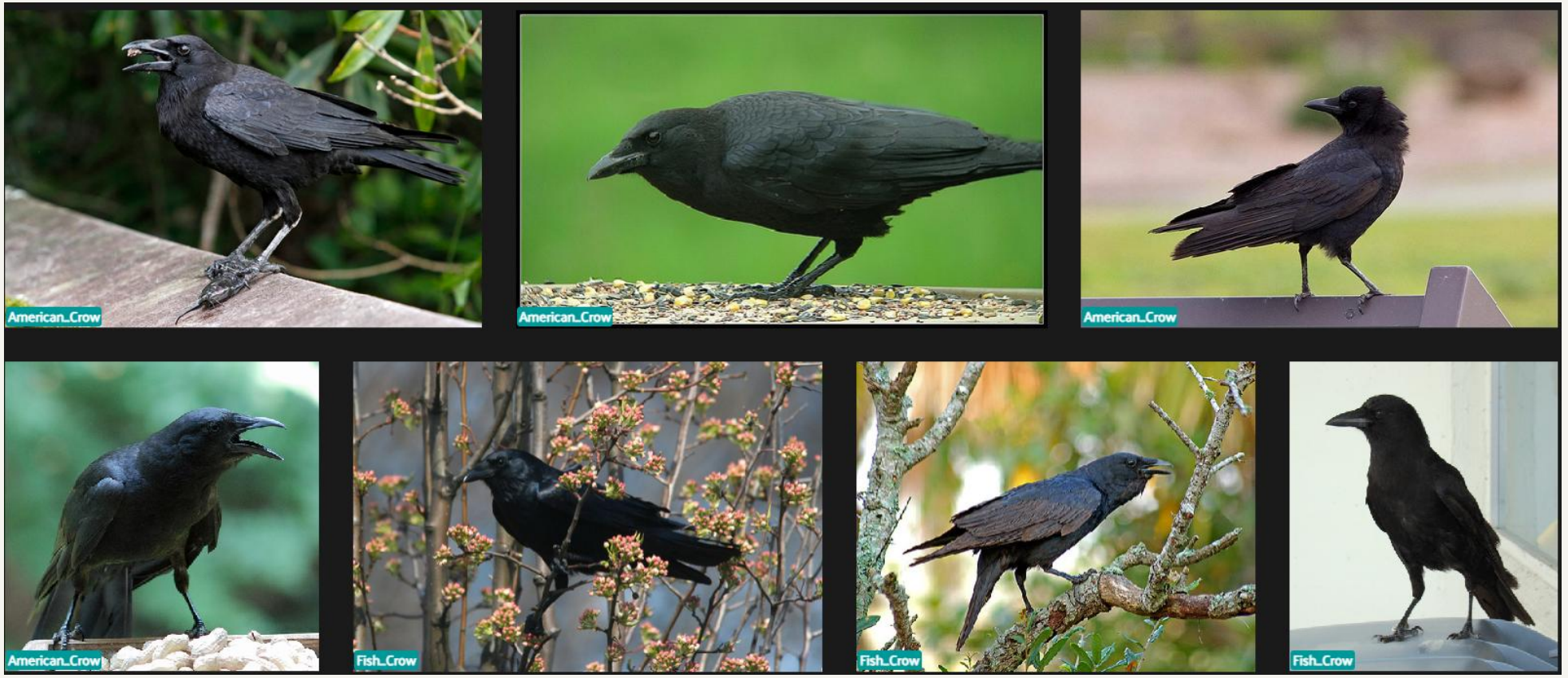
Why?

- **5-fold CV showed systematic confusion for crows**
- **No new manual labels allowed -> try self-training to add samples for the hardest classes**

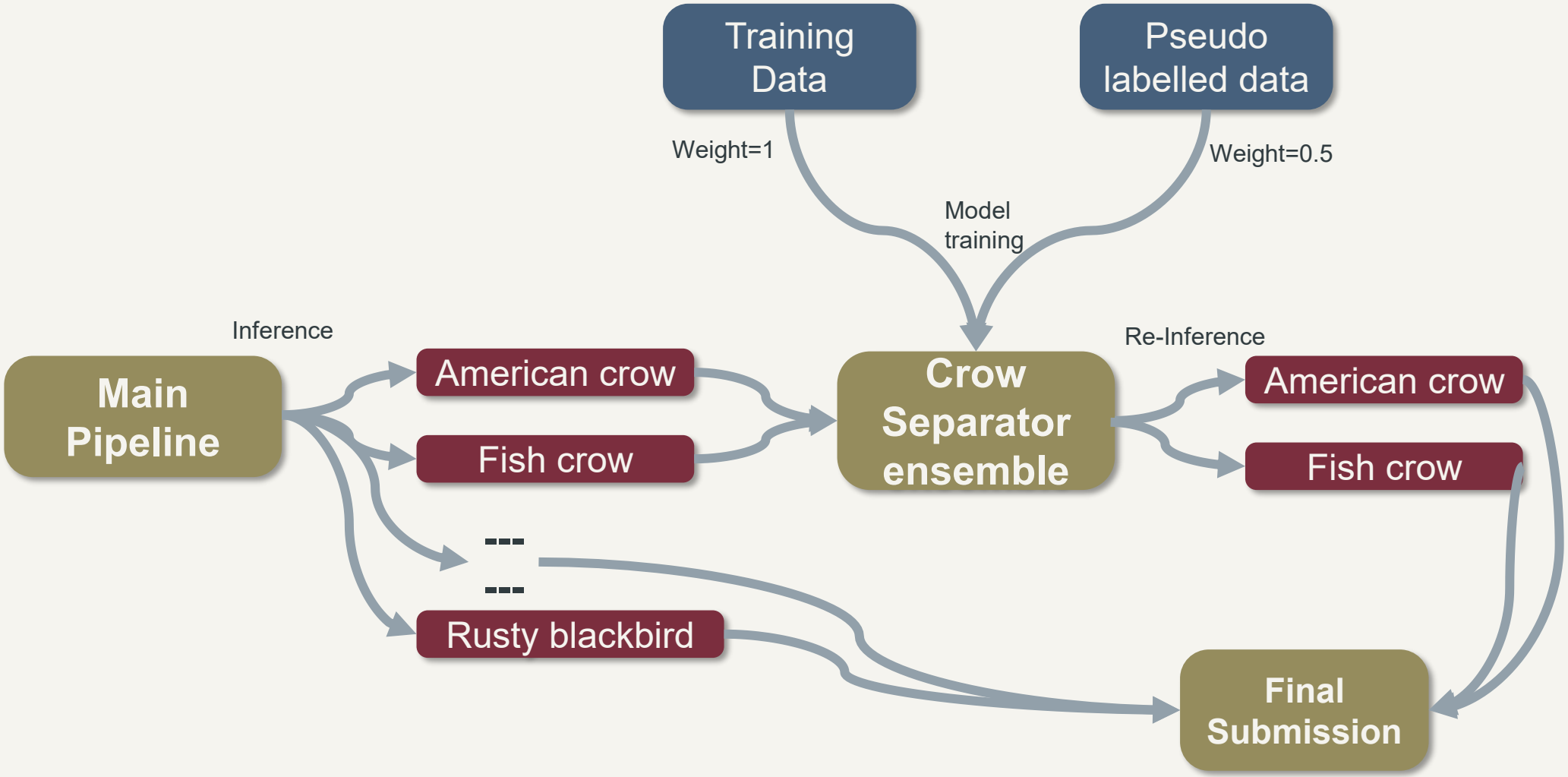


BINARY CROW SEPARATOR

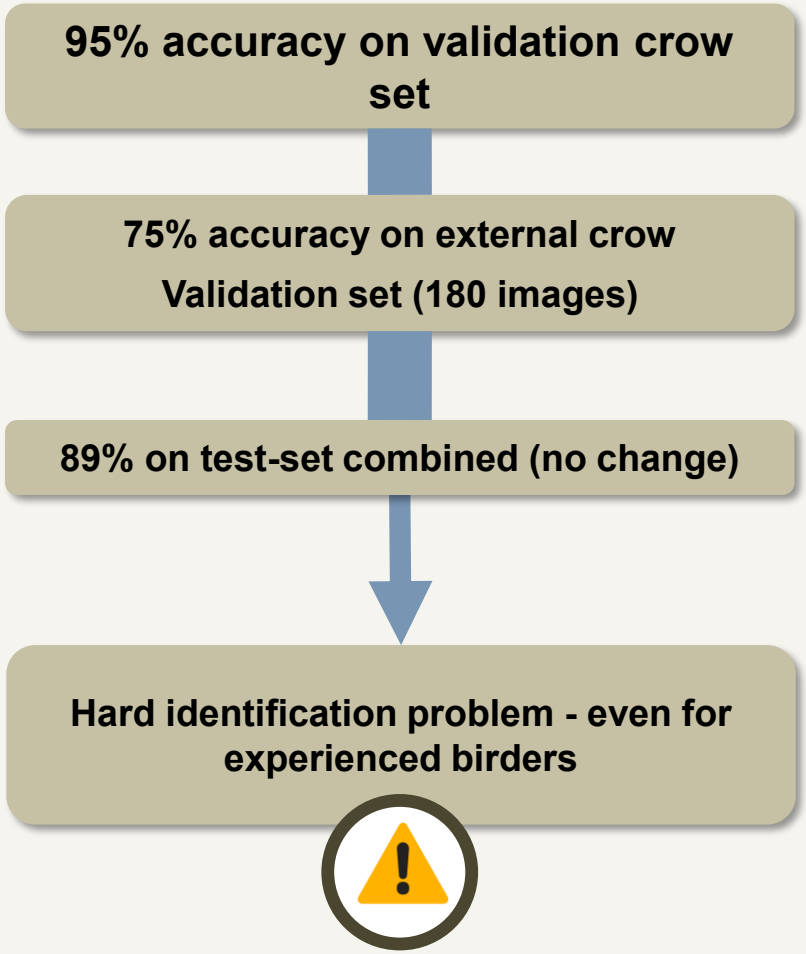
Crow separation: American Crow and Fish Crow are **visually near-identical** in most photos



CROW SEPARATOR PIPELINE



CROW SEPARATOR RESULTS



SUMMARY RESULTS ON TEST SET



Model	Average Test Score	Private Test Score
ViT-384 & linear head	**90%**	90%
ViT-384 & class weighting	89.5%	89%
Eva02 & arcface head	89.25%	85.5%
Eva02 & linear head	88.3%	87%
Eva02, arcface & crow separator	89%	87%
EfficientNet & LogReg	74.8%	77%

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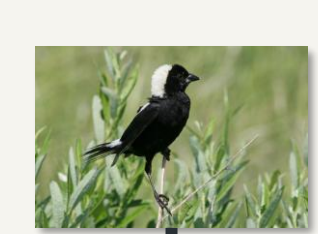
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FINAL PIPELINE



Raw Trian / Val Images



Cropping (GT)







Normalizing + Augmentation

Backbone

Linear Head

ArcFace Head

Class Prediction

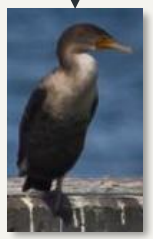
9 (Bobolink)

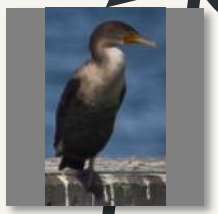
4 (Brandt Cormorant)

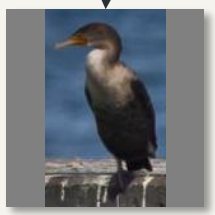
Raw Test Images

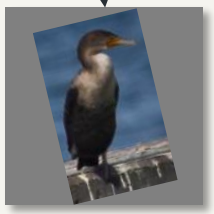


Cropping (R-CNN)









CONCLUSIONS

Selected Model: ArcFace EVA-02 Large

Public: 93%

Private: 85.5%

Best Model: ViT-384 (Linear)

Public: 90%

Private: 90%

Data Quality > Model Complexity: these provided consistent improvements

In fine-grained classification with **limited data** (especially with the way the test data is structured), any **added complexity can easily lead to overfitting**

Achieving close to **100% is impossible**: species pairs indistinguishable, data quality for some samples, and training on males vs. testing on females in some cases

Future: Better pretrain data (within rules)**, better data quality control, "localized" focus**

Occam's Razor: "Entities should not be multiplied without necessity."

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